

SOIL | PLANTS | PEOPLE | A HEALTHY PLANET

When 13 Inches of Rain Falls in Eight Hours

What a PQNK field reveals about hardpan, infiltration and the soil's capacity to receive rain

Abstract

Thirteen inches (330 mm) of rain fell in eight hours. A field video recorded by Jaskaran Singh, a PQNK farmer in Fazilka district, Indian Punjab, shows what happened next. On his PQNK beds, where the hardpan had been broken, the water had gone into the soil and he could walk on the beds in slippers; in the neighbouring conventionally managed orchards and fields, water stood on the surface. In one strip of his own orchard, where beds had been shaped but the hardpan had not been broken, water was still standing too. The rain was the same everywhere; the difference was the hardpan. The lesson is that hardpan is not only a restriction on roots. It is also a barrier to water, and once it is broken and the soil is protected from being compacted again, the land itself becomes the first place where rain is received.

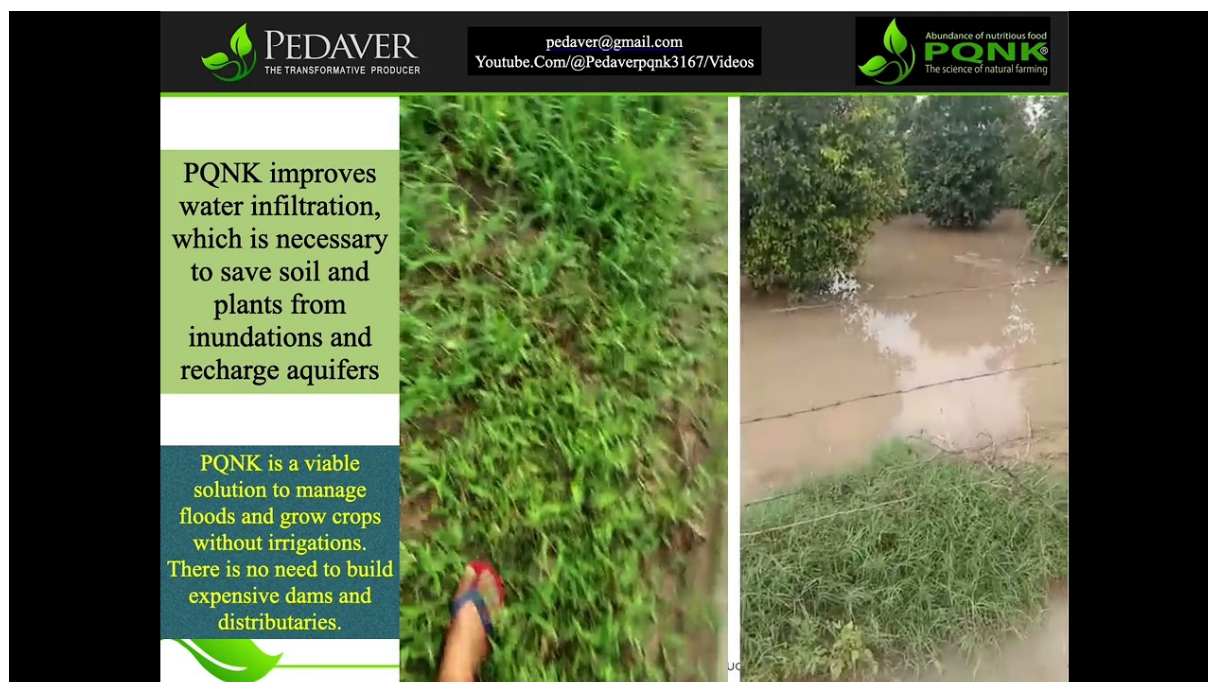
1. The Field Observation

Thirteen inches of rain fell in eight hours on the farm of Jaskaran Singh in Fazilka district, Indian Punjab. He had made PQNK permanent raised beds in his orchard and broken the hardpan. When he walked his field after the storm, the water on the PQNK beds had soaked into the soil within a short time; the beds were firm and covered, and he walked on them in slippers. Across the boundary, the neighbouring orchards and fields still stood under water. Within his own orchard there was a second comparison: a strip where beds had been shaped but the hardpan had not yet been broken was still holding water.



Figure 1. After 13 inches of rain in eight hours: a conventional, flood-irrigated orchard under water (left) and PQNK beds in Jaskaran Singh's orchard, firm enough to walk on in slippers, with water only in the furrow between the beds (right). Stills from his field video, Fazilka district, Indian Punjab.

The rain was the same. The visible difference was the land's capacity to receive it.



2. Hardpan Is Also a Water Problem

Hardpan, or plough pan, is the compacted subsurface layer formed progressively below the normal working depth of repeated tillage and machinery traffic. In PQNK these are not treated as two different phenomena. The same restrictive layer that limits root penetration can also restrict the downward movement of water.

When intense rainfall reaches a field whose effective soil depth is capped by a dense subsurface layer, the upper soil can fill rapidly. Once its immediate storage and transmission capacity is exceeded, water has fewer pathways downward. Ponding, lateral movement and runoff become more likely. The field begins to behave like a shallow container.

Breaking the hardpan therefore has consequences beyond root growth. It reconnects the cultivated surface with a deeper soil profile and creates the opportunity for rainfall to move downward rather than being confined near the surface.

3. Breaking the Layer Is Only the Beginning

PQNK does not regard subsoiling as a recurring cultivation operation. Where a restrictive hardpan exists, it is corrected once, with one subsoiler pass to about 22 inches. The objective is not to replace shallow tillage with repeated deep tillage. It is to remove the inherited physical restriction and then stop rebuilding it.

After correction, the soil is protected through the wider PQNK system: no repeated soil disturbance, permanent biological cover, retained roots and residues, biodiversity, and controlled machinery traffic in the furrows rather than across the protected production bed. Roots, organisms, aggregation and pore continuity are then allowed to develop without being repeatedly reset.

4. From Rainfall to Infiltration

The hydrological sequence can be understood simply. Rain first reaches the land surface. If the surface is protected and the soil profile beneath it contains connected pore space without a restrictive hardpan, water has more opportunity to enter and move through the profile. Some is retained as soil moisture. Some becomes available to roots. Some may move below the active root zone, and, where geology permits, a portion may eventually contribute to groundwater recharge.

Infiltration is the first step and groundwater recharge the last. Water that enters the profile first becomes soil moisture held for the roots; what moves below the root zone continues downward towards the groundwater, as the depth of the soil and the geology allow. The field shows the first step plainly: on the PQNK beds the water left the surface and entered the soil.

5. The Landscape Is the First Reservoir

Conventional water planning often begins after rainfall has already become runoff. Water is collected in ponds, reservoirs and dams. These structures can have legitimate purposes, but they operate downstream of the first hydrological opportunity: the soil directly beneath the point where the raindrop lands.

PQNK moves the first intervention upstream. Before asking how much runoff can be captured, it asks how much rainfall can be received by the land itself. A field with restored infiltration capacity distributes water entry across the treated surface rather than waiting for water to concentrate at a few collection points.

“The first opportunity to harvest rain is the soil pore beneath the point where each raindrop lands.”

6. A Reservoir Captures Runoff. Receptive Soil Can Prevent Some Runoff From Forming.

The distinction is fundamental. A reservoir receives water after rainfall has concentrated and begun moving across the landscape. A receptive soil begins receiving water before that stage. The two approaches are not mutually exclusive, but the sequence matters. Restoring the land's capacity to absorb rainfall should come before assuming that every excess-water problem must be solved by larger downstream storage.

Under very intense rainfall, even healthy soil can reach its infiltration or storage limit, and runoff can still occur. PQNK therefore does not claim that hardpan correction makes flooding impossible. It increases the opportunity for rainfall to enter and be retained within the soil profile before water becomes surface runoff.

7. Why the Comparison Matters for Agriculture

Agricultural hardpan creates a double penalty. During dry periods it restricts roots from exploring deeper moisture and a larger volume of soil. During intense rainfall it restricts downward water movement. The same layer can therefore contribute to both drought vulnerability and waterlogging.

Correcting the restriction and then preserving the soil profile changes that relationship. Deeper rooting increases the volume from which plants can obtain moisture and nutrients. Better-connected pore space increases the opportunity for rainfall to enter the profile. Permanent surface cover further protects the soil from direct raindrop impact, reduces crusting and unnecessary evaporation, and helps preserve the biological and physical structure through which water moves.

8. What This Field Evidence Shows

The video is valuable because it records a real field under an unusually severe rainfall event rather than a laboratory column or a small infiltration ring. It shows crop-scale and field-scale behavior under practical management.

It shows that after 13 inches of rain in eight hours, the PQNK beds with the hardpan broken had taken the water into the soil, while the conventional land next to them, and the strip of the same

orchard where the hardpan had not yet been broken, still held water on the surface. Because the strip with unbroken hardpan sits in the same orchard, under the same rain, the comparison points directly at the hardpan as the barrier.

9. Connection With “Rain Where It Falls”

This field evidence is a practical companion to the PQNK Knowledge Paper “Rain Where It Falls: Restoring the Soil Before Building More Water Storage” (KP-212). That paper develops the landscape-scale argument that rainfall management should begin by restoring the soil’s capacity to receive water where it falls. The present video shows what that principle can look like during an extreme event.

The broader objective is not to remove every dam or pond. It is to correct the order of intervention: first restore the soil and the landscape as the place where rain is received; then decide what genuine surplus runoff remains and what additional storage or flood-management infrastructure is actually required.

Conclusion

Thirteen inches of rain in eight hours is an extreme test of any agricultural landscape. The farmer-recorded comparison shows that land condition can profoundly alter what happens next. Where the hardpan had been properly broken and the soil profile was being managed under PQNK, water infiltrated far more effectively than in the conventional comparison visible in the footage.

The lesson is larger than one storm. Hardpan is not merely an agronomic problem beneath the roots. It is a hydrological barrier. Restoring the continuity of the soil profile, and then protecting that continuity from renewed disturbance and compaction, allows the land itself to become the first place in which rainfall is received, stored and cycled.

PQNK Principle: Before building more places to store runoff, restore the land’s ability to receive the rain.

Field Evidence and Source Note

Field evidence: video “PQNK improves water infiltration”, recorded by Jaskaran Singh, PQNK farmer, Fazilka district, Indian Punjab, after 13 inches of rain in eight hours. It compares his PQNK orchard beds with the neighbouring conventionally managed orchards and fields, and with a strip of his own orchard where the hardpan had not been broken.

Pedaver/PQNK interpretation: hardpan correction, permanently protected soil, biological cover and controlled traffic in the furrows are discussed within the established PQNK production framework.

Related PQNK Sources

1. PQNK Knowledge Paper KP-212: Rain Where It Falls. <https://pedaver.com/papers/rain-where-it-falls/>
2. PQNK Book, Chapter 5: The Natural Water System. <https://pedaver.com/books/natural-ecosystem-science/the-natural-water-system/>
3. PQNK Book, Chapter 25: Hardpan. <https://pedaver.com/books/natural-ecosystem-science/hardpan/>
4. PQNK Book, Chapter 26: Soil Moisture Management. <https://pedaver.com/books/natural-ecosystem-science/soil-moisture-management/>

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